MINING DEALER BEST PRACTICE

Managing Haul Road Defects with MineStar™ Edge & HEI

Maintenance and Repair | **Application** | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

This best practice article serves as a guide on how to utilize Road Analysis Control (RAC) in conjunction with MineStar Health Equipment Insights (HEI) to improve haul road quality. This article also identifies haul road maintenance KPI's that can be utilized to monitor road management effectiveness. An operations ability monitor and improve haul road conditions will directly affect their haul trucks performance by decreasing stress on major components, in turn increasing fleet reliability. Smooth haul roads also allow trucks to increase cycle time, upping material movement.

2.0 Best Practice Description

By utilizing the Road Analysis Control (RAC) system on Caterpillar large OHTs paired with MineStar Health Equipment Insights (HEI), mine operations and maintenance teams can identify the influx, severity, and location of the haul road area that the truck flags as problematic. Both technologies are covered in depth below.

2.1 RAC Overview

Road Analysis Control is an onboard technology used to record and benchmark haul road quality. RAC is integrated with the Vital Information Management System (VIMS™) and measures key data such as impact and shock to the suspension system. This information is communicated to the operator through RAC event codes and to external asset monitoring software such as Health Equipment Insights (HEI) using a radio network.

RAC collects each strut pressure data, utilizing the same sensors that determine payload and cycle time information, to measure frame movement. These sensors are set to measure frame Pitch (front-to-rear shift in weight), Rack (The lateral twisting motion), and Bias (side-to-side movement). If strut pressures exceed a specified threshold for Pitch, Rack, or Bias an event code will be generated.

Haul road conditions must be monitored closely as over time, the stress of rough roads can lead to premature wear to haul truck suspension, steering, frames, and damage to tires and wheels. Poor haul roads also introduce unnecessary stress and strain to operators.

2.2 MineStar Health Equipment Insights (HEI)

MineStar Health Equipment Insights (HEI) is a cloud-based, internet-accessible application that visualizes machine electronic data (VIMS and Product Link), fluid samples, and inspections when CAT inspect is used. HEI allows maintenance staff to diagnose equipment events, monitor machine parameters, and analyze trends. HEI also provides a variety of production data to assist operational personnel in tracking key performance indicators (KPI).

HEI comes equipped with a pre-loaded report that displays the information captured from the RAC system. This report, labeled "Haul Road Condition" allows users to quickly reference the following information:

- RAC event Map & Counts
- Payload Overload Map

- Severity Index KPI
- FELA Trends
- CAT 10-10-20 Payload Summary

3.0 Implementation Steps

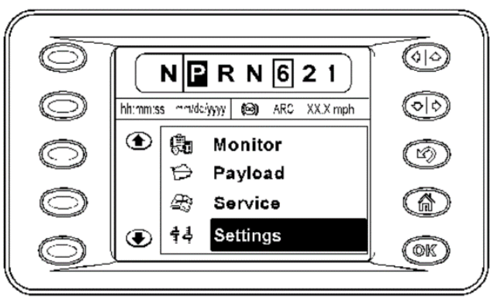
3.1 Action Planning

The following section outlines how to utilize HEI’s RAC event map to develop a work scope for haul road maintenance teams.

3.1.1 Identifying Problem Areas

The first step to managing haul road conditions is to identify and rank problem areas. HEI comes equipped with a Haul Road Condition report that allows users to identify where RAC events are being triggered on a map. Under the “Machine Conditions” tab, users can view the number of RAC events and trigger locations on a per-unit basis. When observing RAC event data, it is important to ensure that each unit in the fleet has been set to the same severity level. RAC severity levels adjust sensitivity thresholds for RAC codes, level 1 being the most sensitive, and level 3 being the least.

RAC severity level can be altered directly through a machines Advisor Display:



1. Select the settings menu
2. Chose the “VIMS” sub-menu
3. Navigate to “RAC Severity”
4. Select a value of 1, 2, or 3

With the Haul Road Map, users can pinpoint the location of all RAC codes in a set date range. In mine sites where there is a large influx of RAC codes, a road maintenance planner may consider shortening the date ranges and filtering the event map to show only medium-level event codes. This will help pinpoint the areas of greatest concern. As haul road quality increases and RAC events frequency decrease, event code levels and date ranges can be extended. In the event that a single haul truck is experiencing significantly more or less RAC codes then the rest of the fleet, the trucks strut pressures and pressure sensors outputs are to be investigated for potential faults.

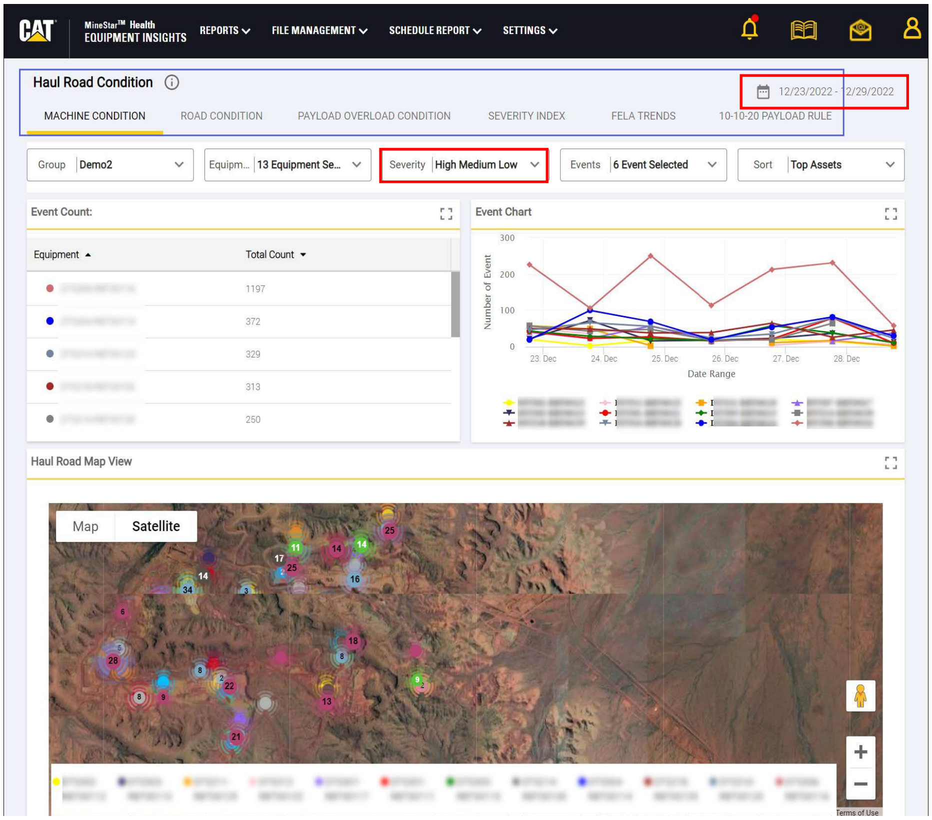


Figure 1 RAC Event Map

3.1.2 Planning repairs

With the use of the RAC event code map, the road maintenance planner can utilize coordinates related to each event to develop and distribute a repair strategy. For mines with a larger footprint, areas of the site can be broken up into sections as seen in Figure 2.

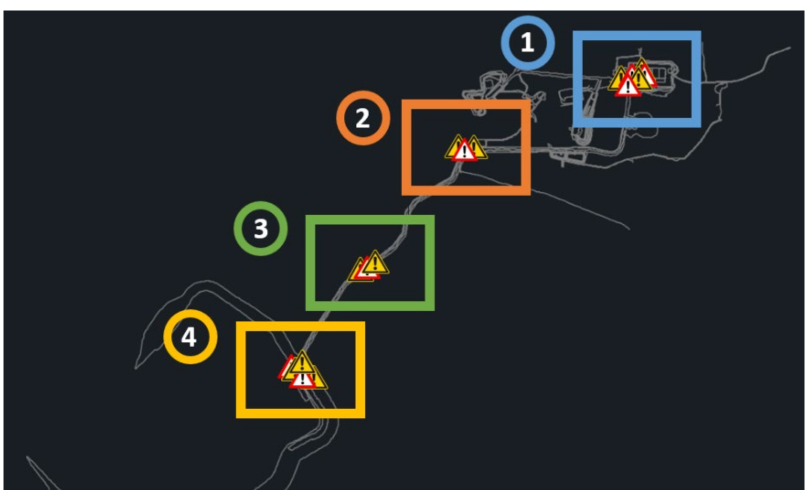


Figure 2 RAC Event Map

It is recommended that repair maps are generated weekly in order to keep up with constantly changing haul road conditions. It is also necessary for road maintenance teams to communicate repair statuses back to the planning team, this data can be stored and used to identify any areas that require constant attention. Additional information on performing RAC analysis is available in the “2.02-210211-1 Road Analysis Control (RAC)” best practice article.

3.1.3 Prioritizing repairs

In mines with troubled haul roads, an influx of RAC codes may make it difficult to decide what routes to prioritize. By utilizing the “Truck Cycles” report generated through Minstar EDGE, a summary can be created that displays what haul roads moved the most material over a given time period:

Dump Area	Tonnage	Tonnage %	% of RAC Events
CRUSHER "A"	181,187	38%	20%
CRUSHER "B"	119,323	25%	24%
WRSA "C"	109,537	23%	44%
STKP "B"	26,588	6%	6%
WRSA "B"	18,822	4%	6%
535-110	17,552	4%	4%

In the example above, the two Dump Areas for “Crusher “A” and “B” show the most tonnage moved, however 44% of the RAC events were triggered on the road to “WRSA “C”. With this information a maintenance team is able to decide to prioritize haul road maintenance based on usage for martial movement or the number of RAC events triggered.

3.2 Monitoring Improvements

To track haul road conditions, there are multiple KPIs that can be generated using HEI. These KPIs will help a mine site trend haul road condition and understand if site conditions are improving.

3.2.1 Severity Index

Severity index is a feature that helps in detecting fluctuation in productivity levels. It does this by taking the number of RAC events (Rack, Pitch, Bias) and dividing this by the number of haul cycles in a day.

This information can be quickly referenced using the “Severity Index” tab. The graph on this tab displays the daily severity index for the selected date range, along with the average over the last 30 days. To trend this data, the road maintenance planner can extract the average severity index and plot it weekly. This information can be trended over time to monitor the effect of haul road maintenance practices. Figure 3 shows a severity index graph of a mine whose haul roads show consistent improvement. When monitoring the severity index, it is important to note that external factors such as a week of bad weather can result in an increased severity index. When reporting haul road KPIs, include a section that notes any factors that may affect haul road data.

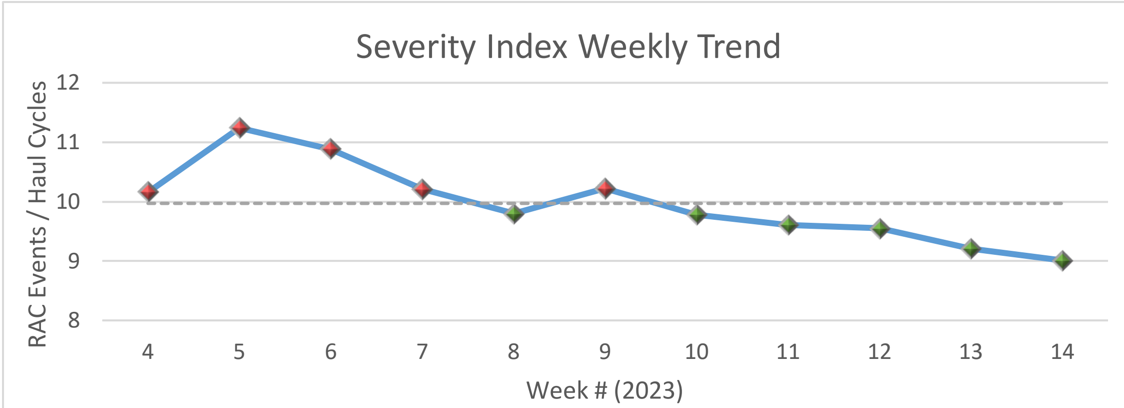


Figure 3 Severity Index Example

3.2.2 Fatigue Equivalent Road Analysis (FELA)

RAC uses FELA to calculate the average Rack and Pitch, producing an index that is used to establish the severity of each cycle. Using the FELA Trend tab in HEI, the user can view the Max Rack and Pitch FELA value along with a trend chart for each unit in the fleet.

Through the dashboard function, users can create a “Trends Beta – Fleet Summary” widget, to view the FELA trends for the entire fleet. This widget also creates a table on the main dashboard screen that displays the MAX and MIN trend values on a per unit basis.

Trends Beta - Fleet Summary Data - -

Channels	Unit		
Truck Payload Cycle Frame Pitch FELA-avg-Road Analysis Control, FELA Good Trigger-161-Communication Gateway #1		Min	1150.00
		Max	1450.00
Truck Payload Cycle Frame Rack FELA-avg-Road Analysis Control, FELA Good Trigger-161-Communication Gateway #1		Min	1400.00
		Max	1600.00

These FELA “Max” and “Min” values can be used to plot a FELA Rack and Pitch range on Caterpillars FELA Curve (KWNE9023) as seen in Figure 4.

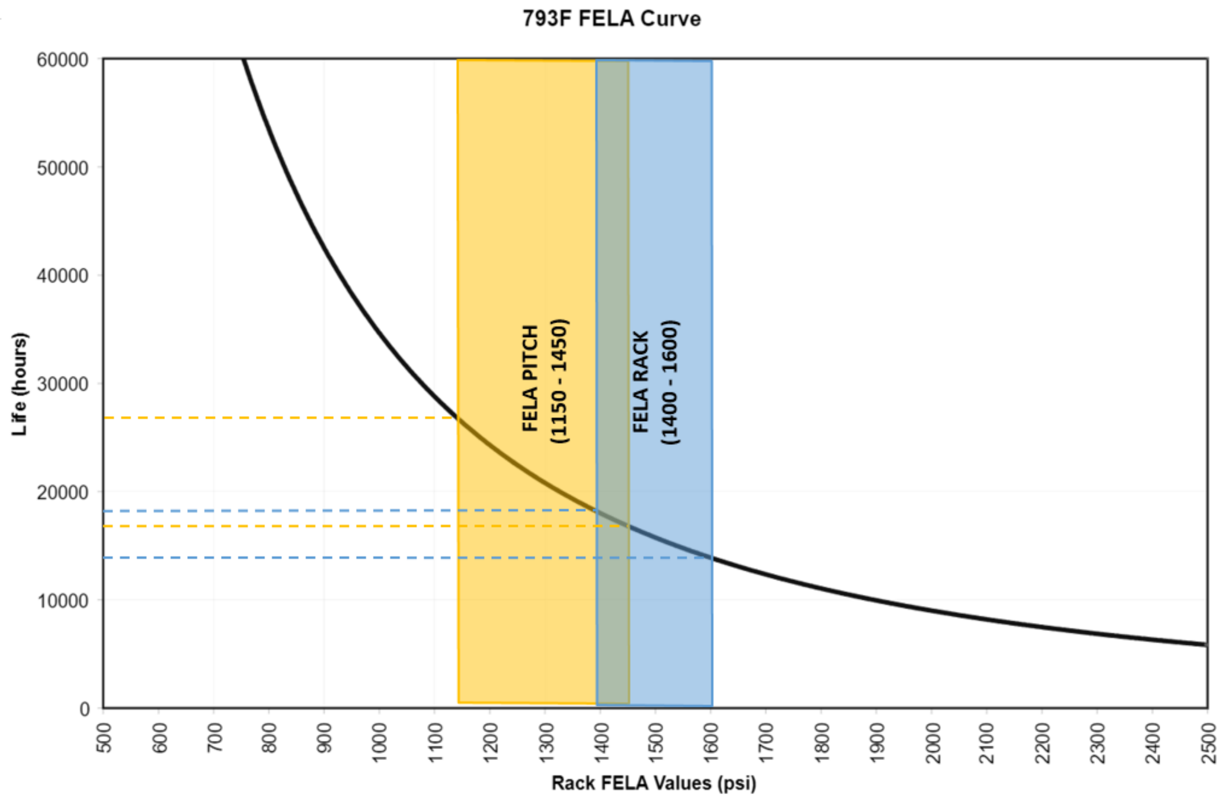


Figure 4 FELA Curve

The purpose of the FELA graph is to help RAC users better understand the relationship between the haul road conditions and the intended design life cycle (machine life). This range can be plotted on a monthly basis in order to track progress of haul road condition and its impacts on equipment life.

3.2.3 Time Series Beta – Datalogger

Minstar HEI allows for the continuous collection of trend data on all subscribed assets. There are multiple data channels that can be used before and after haul road improvements to help gauge the effect on equipment efficiency and productivity.

Some useful data channels include:

- Actual Gear-81 Trans Ctrl
  - Displays gear shift patterns and number of gear shifts in a given cycle
- OHT Payload State 161
  - Displays the current payload state (Traveling Loaded, Traveling Empty, Dumping etc…) to help understand where a truck is in its cycle
- Fuel Consumption Rate
  - Displays the momentary fuel consumption rate of each truck
- Ground speed
  - Displays ground speed of truck

Figure 5 Timeseries Beta - Datalogger

Figure 5 displays a Datalogger graph for a complete haul cycle of a 793F, with all channels above being displayed. HEI stores all Datalogger information, allowing the user to compare truck performance before and after haul road improvements have been made. The “4.0 Benefits” section displays an example where Datalogger information was used to validate 793F haul cycle improvements.

4.0 Benefits

Combining RAC with HEI allows mine sites to efficiently identify problem areas and repair haul roads. If a site is able to utilize these programs to reduce the severity level of their haul fleet from 11 to 7, the fleets reliability will be significantly improved. The graphs below (Fig. 6) show an actual example of a 793F as it approaches a rough area in the haul road. During this snapshot the haul truck undergoes three gear changes, decelerates from 48KM/H to 35KM/H, and accelerating again from 35KM/H to 45KM/H. Three “Level 1” and One “Level 2” RAC events are triggered over this area of road.

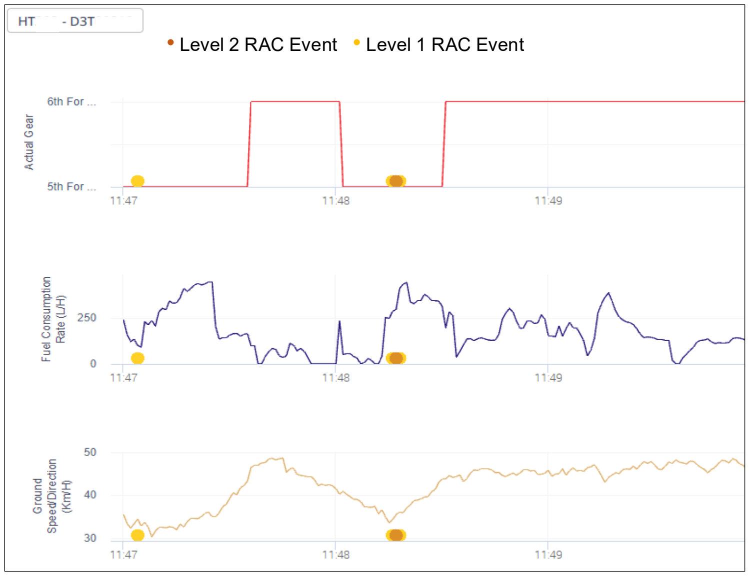


Figure 6 RAC Data Log 1

The snapshot below “Fig. 7”, demonstrates the same area of haul road after repairs. During this event the truck underwent no gear changes, and its speed remained constant.

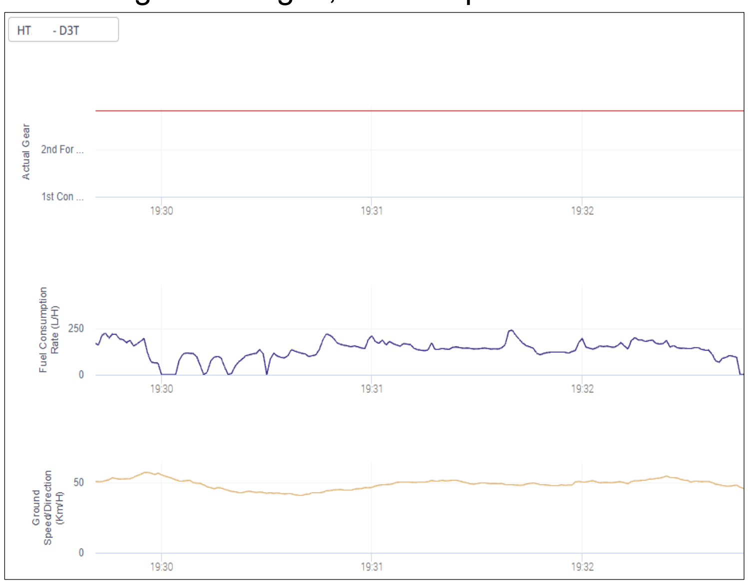


Figure 7 RAC Data Log 2

In the event that an operations team is able to repair and maintain four haul road areas, yielding similar results, the 793F will experience a reduction in:

- Haul Cycle Time
  - Speeds remained constant, no time loss undergoing acceleration and deceleration
  - Haul cycle time reduces by 5%
- Fuel Burn
  - Fig. 5 displays spikes in fuel consumption from 100 L/HR to 450 L/HR, while Fig. 6 fuel burn rates stay between 100 L/HR to 250 L/HR

    - The elimination of these spikes will reduce the Avg. fuel burn rates by up to 5% per cycle
- Major Component Wear
    - Lower FELA values will result in suspension strut longevity
    - ~12 Less gear changes per cycle reduces wear on transmission and powertrain
- Tire Consumption
    - Elimination of four haul road areas with increased risk to tire damage
- Operator Stress
    - Reduced operator fatigue
    - Increased haul road safety

5.0 Resources Required

- VIMS ABL or Later
- RAC Installed on all Large Off-Highway Trucks (REHS0875)
- MineStar Health Equipment Insights – Subscription and Activation on all OHTs
- Update Haul Road Maps
- Haul Road Maintenance Personnel & Equipment

6.0 Supporting Attachments / References

- FELA Graph
    - Doc. KENR9023
- RAC Additional Information
    - Doc. RENR2639, RENR2636

7.0 Related Best Practices

0409-2.02-1165	Haul Road Optimization
1106-1.1-1035	Cost Reduction Through Haul Road Management Process
2.02-210211-1	Road Analysis Control (RAC)
307-2.05-1320	Using Video Overlay for Communication of Mine Site Opportunities

8.0 Acknowledgements

This Best Practice was written by:

Riley Yetman
Mining Performance Solutions Analyst – Toromont CAT
Ryetman@toromont.com